# Climbing the Clifford hierarchy

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## Defining the hierarchy

Pauli matrices:

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \ X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \ Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \ Y = iXZ.$$

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Pauli group on $n$-qubits:

$$\mathcal{P}_n = \mathcal{P}_1^{\otimes n} = \{P_1 \otimes P_2 \otimes \cdots \otimes P_n \mid P_i \in \mathcal{P}_1\}.$$

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The **Clifford hierarchy** on $n$-qubits:

$$\mathcal{C}_n = \bigcup_{k=1}^{\infty} \mathcal{C}_n^{(k)}$$

where

$$\begin{aligned} \mathcal{C}_n^{(1)} &= \mathcal{P}_n, \\ \mathcal{C}_n^{(k)} &= \{ U \in \mathbb{U}(2^n) \mid U\mathcal{P}_n U^\dagger \subset \mathcal{C}_n^{(k-1)} \}. \end{aligned}$$

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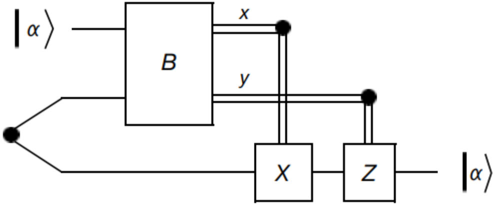
The **Clifford group** on $n$-qubits:

$$\mathcal{C}_n^{(2)} = \{U \in \mathbb{U}(2^n) \mid U\mathcal{P}_n U^\dagger \subset \mathcal{P}_n\}.$$

# Why the Clifford hierarchy (Gottesman & Chuang 1999)

Quantum Teleportation:



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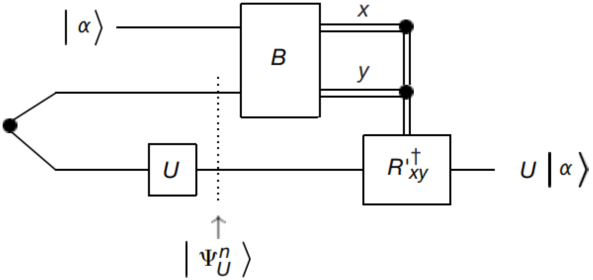
"Erroneous" Quantum Teleportation:



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(2) For universal computation, non-Clifford gates are needed.

(3) The level of the hierarchy lower bounds the implementation complexity.

- Trade-off: more less complex gates vs less more complex gates

## What's known: Part I

The second level, aka the Clifford group, is generated by

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For instance $C = (E_1 + E_2 + E_3 + E_4)/2$, where

$E_1 = I \otimes Y \otimes I, \quad E_2 = I \otimes X \otimes Y, \quad E_3 = Z \otimes Z \otimes Y, \quad E_4 = X \otimes Z \otimes Y$

is unitary but does not belong to the hierarchy.

# What's known: Part II

The diagonal part of each level

$$\mathrm{d}\mathcal{C}_n^{(k)} = \{C \in \mathcal{C}_n^{(k)} \mid C \text{ diagonal}\}$$

is a group and is generated by diagonal matrices

$$Z^{1/2^{k-1}} = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}, CZ^{1/2^{k-2}} = \begin{bmatrix} I & 0 \\ 0 & Z^{1/2^{k-2}} \end{bmatrix}, \cdots, C^{(k-1)}Z$$

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Note that entries are $2^{k-1}$ roots of unity, and they can be determined by polynomials (with $n$ variables) of degree at most $k$.

# What's known: Part III

**Definitions**:

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**Theorem**:

(1) Every matrix in $\mathcal{C}_n^{(k)}$ is semi-Clifford when $n = 1, 2, \forall k$ and $n = 3, k = 3$.
(2) Every matrix in $\mathcal{C}_n^{(3)}$ is generalized semi-Clifford.

# Back to the Clifford group

**Definition**: If $E \in \mathcal{P}_n$ is a Hermitian Pauli matrix ($E^\dagger = E$), then a **Clifford transvection** is a matrix of the form

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$C$ is supported on a subgroup of $\mathcal{P}_n$. If $C$ is Hermitian Clifford, then one can show that $S$ is abelian.

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Let $U \in \mathbb{U}(2^n)$ be Hermitian and let

$$\widehat{U} = (I + iU)/\sqrt{2}.$$

**Is it true that**

$$U \in \mathcal{C}_n^{(k)} \implies \widehat{U} \in \mathcal{C}_n^{(k+1)}?$$

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(5) He have partial results on what matrices from $\mathcal{C}^{(2)}$ climb to $\mathcal{C}^{(3)}$.

Thank You!